Higher Colleges of Technology

كليات التقنية العليا

CDS 2413- PROGRAMMING FOR DATA SCIENCE

Week 2

Week 2	CLO 1: Demonstrate an understanding of data analytics principles to solve business problems

Document Revision CONTROL (DRC)

Version	Author	Effective Date	Change Description	DRC No
1.0	Mohammed Abdul Rahim	2/05/2023	Define the first version	001

X.Y
X: Major change
Y: minor change

Lecture Outline

- Data Wrangling and its Importance
- Data Loading
- Sampling
- Data Cleaning
- Data Projection

LECTURE OBJECTIVES

- At the conclusion of the lecture, the student should be able to perform the tasks below:
  - Understand the significance of data wrangling
  - Understand the purpose of data cleaning
  - Capable of examining the intent of sampling
  - Learn how to write Python code for data projections

Data Wrangling

Data Wrangling - Quiz

Identify potential data quality issues or areas for improvement.

Customer_ID	Review_Date	Review_Text	Rating	Purchase_Amount	Product_ID	Age	Location
C001	01/02/2023	"Great product!"	5	45.99	P123	29	New York
C002	NaN	"Loved it, but arrived late."	4	NaN	P124	NaN	NaN
NaN	02/15/2023	"Didn't like the quality"	2	20.50	NaN	34	Los Angeles
C003	02/10/2023	"Fantastic service and good quality!"	5	59.00	P125	42	Chicago

Data Wrangling - Quiz

Data after Wrangling

Customer_ID	Review_Date	Review_Text	Rating	Purchase_Amount	Product_ID	Age	Location
C001	01/02/2023	"Great product!"	5	45.99	P123	29	New York
C002	01/25/2023	"Loved it, but arrived late."	4	50.00	P124	30	Unknown
C005	02/15/2023	"Didn't like the quality"	2	20.50	P126	34	Los Angeles
C003	02/10/2023	"Fantastic service and good quality!"	5	59.00	P125	42	Chicago

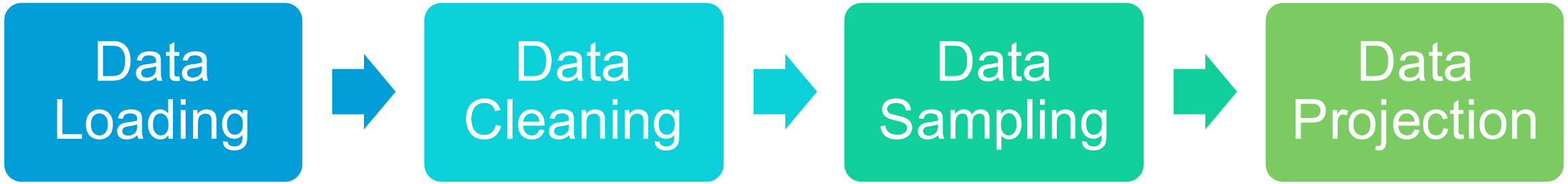
Data Wrangling

- The initial phases of the data analytics process are often referred to as Data Wrangling.
- Data Wrangling, also known as data cleaning, is the process of transforming raw data into a usable format.
- It ensures that the data is accurate, consistent, and ready for analysis.
- It also ensures data quality, improves data usability for decision-making, and reduces errors.

Importance of Data Wrangling

- **Improves Data Quality**
  - Helps to identify and correct errors, missing values, and inconsistencies in raw data.
- **Increases Data Usability**
  - Compatible with the analysis tools and techniques that will be used.
- **Enables Data Discovery**
  - Reveal patterns, anomalies, and relationships in the data.
- **Supports Decision-Making**
  - Provides the foundation for accurate and reliable data analysis to supports better decision-making.
- **Reduces Errors**
  - Helps to prevent errors from being introduced into the data analysis process.

Steps in Data Wrangling



Data Loading

Data Loading

- Data loading is the process of bringing data into a Python environment for analysis and manipulation.
- Python offers several libraries and tools to efficiently load data from various sources, including files, databases, and APIs.
- NumPy and Pandas are popular libraries for numerical and data analysis

Data Loading

- Some important libraries used in python for data analysis are:

The syntax for importing library :

```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
```

Library	Usage
Numpy	for scientific computing and performing basic and advanced array operations.
Pandas	help developers work with "labeled" and "relational" data.
Scikit-Learn	Data scientists use it for handling standard machine learning and data mining tasks such as clustering, regression, model selection, dimensionality reduction, and classification.

Data Loading using Excel File

Pandas allows you to access various sources of data including Excel, Comma Separated Values (CSV), tables, text, database, web and more.

The general syntax to read data from excel file is as follows:

Pandas

The targeted sheet.

Variable = **pd**.read_excel(”Filename”, sheet_name=”Sheet1”)

Any variable name that you create to store the **DataFrame.**

The function that reads data

The file name including extension.

Data Loading using Excel File

The file name with the full extension.

This line imports pandas and create an object pd

```
import pandas as pd

data1 = pd.read_excel("../Data/HigherEducationOpenData.xlsx",
                      sheet_name="Census 2015-2016")
```

Variable name given to **DataFrame** that store the data read from excel file.

The name of the targeted sheet.

	Unnamed: 0	Enrolled and Graduated students by Higher Education Institution , Specialisation and Level of Education	Unnamed: 2	Unnamed: 3	Unnamed: 4	Unnamed: 5	Unnamed: 6	Unnamed: 7	Unnamed: 8
0	NaN	Census 2015-2016	NaN	NaN	NaN	NaN	NaN	NaN	NaN
1	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
2	NaN	Higher Education Institution	Specialisation	Enrolled _ UnderGraduate	Enrolled_Post Graduate	Graduate _ UnderGraduate	Graduated_Post Graduate	NaN	Back to Contents\n<--------
3	NaN	American University in Dubai	Architecture and Construction	287	25	47	3	NaN	NaN
4	NaN	American University in Dubai	Business	979	72	207	22	NaN	NaN

Note: filename and work sheetname are case sensitive.

Data Loading using CSV File

Read data from a flat file - Comma Separated Values (CSV).

Variable = **pd**.read_csv(”Filename.csv”, delimiter=‘,')

Example

Data/uae.csv

```
Abu Dhabi,2 Dec 1971,2784490
Ajman,2 Dec 1971,504846
Dubai,2 Dec 1971,4177059
Fujairah,2 Dec 1971,152000
Ras Al Khaimah,10 Feb 1972,416600
Sharjah,2 Dec 1971,2374132
Umm Al Quwain,2 Dec 1971,72000
```

```
emirates = pd.read_csv("uae.csv",delimiter = ',')
```

```
emirates
```

	Abu Dhabi	2 Dec 1971	2784490
0	Ajman	2 Dec 1971	504846
1	Dubai	2 Dec 1971	4177059
2	Fujairah	2 Dec 1971	152000
3	Ras Al Khaimah	10 Feb 1972	416600
4	Sharjah	2 Dec 1971	2374132
5	Umm Al Quwain	2 Dec 1971	72000

Data Loading using HTML File

Variable = **pd**.read_html(”url”)

Population [edit]

The United Arab Emirates witnessed significant population increase during the past few years because of major growth in the various eco influx of workers from diverse cultural and religious background. From 4.1 million in 2005 to roughly 9.5 million in 2018.[5] Out of this, the r around 11.5% and the remaining 88.5% made up of Expatriate workers. The largest group of non-UAE nationals are South Asian 59.4% (Bangladeshi 9.5%, Pakistani 9.4%, others 2.3%), Egyptian 10.2%, Filipino 6.1%, other 12.8%.[6]

Female citizens and non-citizens account for 28% percent of the UAE'S population due to the high level of male foreign workers.[7] The m population falls in the age group of 25 to 54 year old. A large part of this can be attributed to the expatriate population of working men and category.[8] Population is heavily concentrated to the northeast on the Musandam Peninsula, the three largest Emirates - Abu Dhabi (2.9 r and Sharjah (1.4 million), are home to nearly 75% of the population.[9][10][11]

Emirate	Census 1975	Census 1985	Census 1995	Census 2005	Est. 2010
Abu Dhabi	211,812	566,036	942,463	1,399,484	
Dubai	183,187	370,788	689,420	1,321,453	
Sharjah	78,790	228,317	402,792	793,573	
Ajman	16,690	54,546	121,491	206,997	
Umm Al-Quwain	6,908	19,285	35,361	49,159	
Ras Al-Khaimah	43,845	96,578	143,334	210,063	
Fujairah	16,655	43,753	76,180	125,698	
Total	**557,887**	**1,379,303**	**2,411,041**	**4,106,427**	**7,512,000 (UN)[12] 8,190,000 (NBS)[13]**

Population pyramid data [edit]

```
url = 'https://en.wikipedia.org/wiki/Demographics_of_the_United_Arab_Emirates'
```

```
dfs = pd.read_html(url)
```

```
dfs[3] # fourth table in the webpage
```

	0	1	2	3	4	5
0	Emirate	Census 1975	Census 1985	Census 1995	Census 2005	Est. 2010
1	Abu Dhabi	211812	566036	942463	1399484	NaN
2	Dubai	183187	370788	689420	1321453	NaN
3	Sharjah	78790	228317	402792	793573	NaN
4	Ajman	16690	54546	121491	206997	NaN
5	Umm Al-Quwain	6908	19285	35361	49159	NaN
6	Ras Al-Khaimah	43845	96578	143334	210063	NaN
7	Fujairah	16655	43753	76180	125698	NaN
8	Total	557887	1379303	2411041	4106427	7,512,000 (UN)[12]8,190,000 (NBS)[13]

Note: read_html returns a list of DataFrame objects

DataFrame

DataFrame: DataFrame is a two (or more) dimensional data structure – basically a table with rows and columns. The columns have names and the rows have indexes.

These are the Columns

index

	Unnamed: 0	Enrolled and Graduated students by Higher Education Institution , Specialisation and Level of Education	Unnamed: 2	Unnamed: 3	Unnamed: 4	Unnamed: 5	Unnamed: 6	Unnamed: 7	Unnamed: 8
0	NaN	Census 2015-2016	NaN	NaN	NaN	NaN	NaN	NaN	NaN
1	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
2	NaN	Higher Education Institution	Specialisation	Enrolled _ UnderGraduate	Enrolled_Post Graduate	Graduate _ UnderGraduate	Graduated_Post Graduate	NaN	Back to Contents\n<--------
3	NaN	American University in Dubai	Architecture and Construction	287	25	47	3	NaN	NaN
4	NaN	American University in Dubai	Business	979	72	207	22	NaN	NaN

Rows

Data Cleaning

Data Cleaning

How to handle when data is missing?

Note: Usually, data missing is represented as NaN

1. Drop missing observations

This will delete all rows that contain any missing data.

```
dataset = dataset.dropna()
```

2. Drop rows where all cells in that row is NA

This will delete rows that are missing all its data.

```
dataset = dataset.dropna(how='all')
```

3. Drop column if they only contain missing values

```
dataset.dropna(axis=1, how='all')
```

4. Fill in missing data with zeros

This will replace NaN with 0.

```
dataset.fillna(0)
```

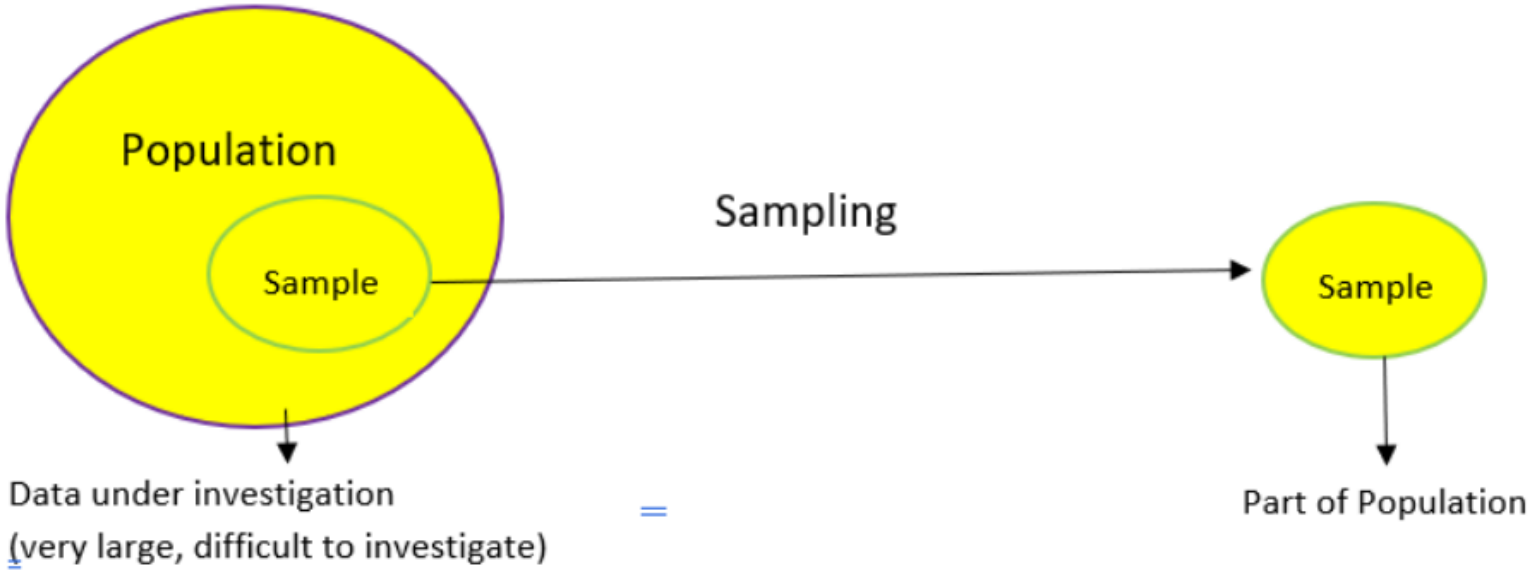
Note:

keyword how="all" to indicate all cells have missing data , keyword how="any" to indicate any cell have missing data

Sampling

Data Sampling(أخذ العينات)

- Sampling is the process of obtaining a small set of data from a large set of data(population)



Data Sampling(أخذ العينات)

- Why sampling is required?
  - It takes less time to choose a sample than it does to select every item in a population.
  - Sample selection is a low-cost technique.
  - A sample analysis is less time-consuming and practicable than a population-wide study.

Data Sampling – Case Study

- The energy is generated from the power plant (Power_Plant.csv) using the following parameters.
  - Temperature
  - Vaccum
  - Pressure
  - Relative Humidity
- Power_Plant.csv contains 9569 records.
- Apply various sampling techniques to obtain the samples from 9569 records for further analysis

Power_Plant.csv

Temperature	Vaccum	Pressure	Relative_humidity	Energy
14.96	41.76	1024.07	73.17	463.26
25.18	62.96	1020.04	59.08	444.37
5.11	39.4	1012.16	92.14	488.56
20.86	57.32	1010.24	76.64	446.48
10.82	37.5	1009.23	96.62	473.9
26.27	59.44	1012.23	58.77	443.67
15.89	43.96	1014.02	75.24	467.35
9.48	44.71	1019.12	66.43	478.42
14.64	45	1021.78	41.25	475.98

Types of Sampling - Random

- Random Sampling: Chosen randomly(عشوائي)

Temperature	Vaccum	Pressure	Relative_humidity	Energy
14.96	41.76	1024.07	73.17	463.26
25.18	62.96	1020.04	59.08	444.37
5.11	39.4	1012.16	92.14	488.56
20.86	57.32	1010.24	76.64	446.48
10.82	37.5	1009.23	96.62	473.9
26.27	59.44	1012.23	58.77	443.67
15.89	43.96	1014.02	75.24	467.35
9.48	44.71	1019.12	66.43	478.42
14.64	45	1021.78	41.25	475.98

Temperature	Vaccum	Pressure	Relative_humidity	Energy
14.96	41.76	1024.07	73.17	463.26
10.82	37.5	1009.23	96.62	473.9
15.89	43.96	1014.02	75.24	467.35
14.64	45	1021.78	41.25	475.98

Types of Sampling – Random - Example

- Create a random sampling of size 100 for Power_Plant.csv

```
# Random Sampling

import pandas as pd
import numpy as np

# Read data from a CSV file
data = pd.read_csv('Power_Plant.csv')

# Generate a random sample of the data
sample_size = 100
sample = data.sample(n=sample_size, replace=False, random_state=42)
```

Some integer number to produce random number except 0 or 1.

Types of Sampling – Systematic

- Systematic Sampling: Chosen based on some condition

Temperature	Vaccum	Pressure	Relative_humidity	Energy
14.96	41.76	1024.07	73.17	463.26
25.18	62.96	1020.04	59.08	444.37
5.11	39.4	1012.16	92.14	488.56
20.86	57.32	1010.24	76.64	446.48
10.82	37.5	1009.23	96.62	473.9
26.27	59.44	1012.23	58.77	443.67
15.89	43.96	1014.02	75.24	467.35
9.48	44.71	1019.12	66.43	478.42
14.64	45	1021.78	41.25	475.98

Temperature > 25.00

Temperature	Vaccum	Pressure	Relative_humidity	Energy
25.18	62.96	1020.04	59.08	444.37
26.27	59.44	1012.23	58.77	443.67

Types of Sampling – Systematic - Example

- Create a systematic sampling by extracting every 5th row from the population to make sample in Power Plant.csv

condition

```
# Systematic Sampling

import pandas as pd

# Read data from a CSV file
data = pd.read_csv('Power_Plant.csv')

# Obtain sample by selecting every 5th row
sample = data.iloc[::5]
```

All columns

Every 5th row

Data Projection

Data Projection

- Data Projection refers to the selection of specific columns (attributes) from a table or query result.
- Purpose
  - Focus on specific information
  - Avoid unnecessary information to improve the performance
  - Simplify data visualization
- Implementation
  - Through various commands such as Dataframe.query, Dataframe.loc, Dataframe.iloc and so on

Data Projection - Dataset

- Data projection is explained in the following slides using following Marks.xls

ID	Division	Quiz	Practical	Project	Final
H00247	EDU	84	67	69.6	67
H00258	HLT	74	68	67	73
H00269	ENG	88	86	75.6	74
H00224	EDU		71	65.8	71
H00269	EDU	22	65	66.8	78
H00277	ENG	40	68	76.4	68
H00277	HLT	71		64.6	70
H00293	BUS	50	74	60.6	78
H00247	CIS	61	70	66.4	82
H00149	APP	97	98	85.2	87
H00295	EDU	70	76	81	84
H00284	CIS	48	64	81.4	68
H00307	ENG	67	77	73	83
H00293	APP	38		64.6	77
H00293	BUS	80	76	68.4	83

Data Projection

```python
# import the dataset
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
```

```python
#Importing the dataset
grades=pd.read_excel("Marks.xlsx",sheet_name="Grades",skiprows=2)
grades
```

	ID	Division	Quiz	Practical	Project	Final	Total
0	H00247	EDU	84.0	67.0	69.6	67.0	NaN
1	H00258	HLT	74.0	68.0	67.0	73.0	282.0
2	H00269	ENG	88.0	86.0	75.6	74.0	NaN
3	H00224	EDU	NaN	71.0	65.8	71.0	NaN
4	H00269	EDU	22.0	65.0	66.8	78.0	NaN
5	H00277	ENG	40.0	68.0	76.4	68.0	NaN
6	H00277	HLT	71.0	NaN	64.6	70.0	NaN
7	H00293	BUS	50.0	74.0	60.6	78.0	NaN
8	H00247	CIS	61.0	70.0	66.4	82.0	NaN

Data Projection

```
#To insert the mean value of each column into its missing rows:
grades.fillna(grades.mean(numeric_only=True).round(1), inplace=True)
print(grades)
```

ID	Division	Quiz	Practical	Project	Final	Total
H00247	EDU	84.0	67.0	69.6	67.0	294.5
H00258	HLT	74.0	68.0	67.0	73.0	282.0
H00269	ENG	88.0	86.0	75.6	74.0	294.5
H00224	EDU	65.4	71.0	65.8	71.0	294.5
H00269	EDU	22.0	65.0	66.8	78.0	294.5
H00277	ENG	40.0	68.0	76.4	68.0	294.5

Data Projection

```
#Display the number of rows and columns
grades.shape
```

```
(24, 7)
```

```
# Display all the columns
grades.columns
```

```
Index(['ID ', 'Division', 'Quiz', 'Practical', 'Project', 'Final', 'Total'], dtype='object')
```

```
# Drop the total column
grades.drop(['Total'], axis=1)
```

Data Projection

```
# Delete the record which are blanks
grades=grades.dropna(how="all")
grades
```

```
# Rename the column Final to FWA
grades.rename(columns={"ID ":"ID","Final":"FWA"},inplace=True)
grades
```

```
# Find the row which has no student id and delete the row
grades = grades.drop(17)
grades
```

Data Projection

```
# Display number of divisions without any duplicate values
len(grades['Division'].unique())
```

```
# Display the first few rows only – default 5 rows are displayed
grades.head()
```

```
# Display the last 3 rows only
grades.tail(3)
```

```
#Display all the rows from 5 to 10
grades[5:11]
```

Data Projection

```
# Using loc command display all the rows from 11 to 20 and columns Quiz to Project
grades.loc[11:20,'Quiz':'Project']
```

```
#Using iloc command display all the rows from 15 to end and 3rd to 5th column
grades.iloc[15:,2:5]
```

```
# Display the columns ID and FWA
grades[['ID','FWA']]
```

Data Projection

```
# Display the student who got more than 69 in Final Marks
grades[grades['FWA'] > 69]
```

```
# Display the student who got Final Marks between 80 to 89
grades[(grades['FWA'] >= 80) & (grades['FWA'] <= 89)]
```

```
# Display the student who study in ENG division and got FWA less than 71
grades[(grades['Division'] == 'ENG' )& (grades['FWA'] < 71)]
```

```
# Display the student sort by Division
grades.sort_values('Division')
```

Data Projection

```
# Display the student sort by Division in ascending and Final Marks by descending
grades.sort_values(['Division','FWA'],ascending=[True,False])
```

```
# Display the number of students in each division
grades.groupby('Division')['ID'].count()
```

```
Division
APP    4
BUS    4
CIS    3
EDU    4
ENG    5
HLT    3
```

Data Projection

```
# Change the index to ID permanently and display the data frame
grades.set_index('ID',inplace=True)
grades
```

```
# Reset the index back to original permanently. Verify it by display the data frame
grades.reset_index(inplace=True)
grades
```

Data Projection

```
#Display the average FWA marks
grades['FWA'].mean()
```

```
75.69565217391305
```

```
# Return the sum of values across each row
grades['Total']=grades[['Quiz','Practical','Project','FWA']].sum(axis=1)
grades
```

	ID	Division	Quiz	Practical	Project	FWA	Total
0	H00247	EDU	84.0	67.0	69.6	67.0	287.6
1	H00258	HLT	74.0	68.0	67.0	73.0	282.0
2	H00269	ENG	88.0	86.0	75.6	74.0	323.6

Data Projection

```
# Display the Statistics Summary for the column Project
grades['Project'].describe()
```

```
#Create a new column 'Final Marks' which is sum of 20% of Quiz, 25% of Practical, 25% of Project and 30% of FWA. Ver
grades['Final Marks'] = grades['Quiz'] * 0.2 + grades['Practical'] * 0.25 + grades['Project'] * 0.25 + grades['FWA']
grades
```

	ID	Division	Quiz	Practical	Project	FWA	Total	Final Marks
0	H00247	EDU	84.0	67.0	69.6	67.0	287.6	71.05
1	H00258	HLT	74.0	68.0	67.0	73.0	282.0	70.45
2	H00269	ENG	88.0	86.0	75.6	74.0	323.6	80.20
3	H00224	EDU	65.4	71.0	65.8	71.0	273.2	68.58
4	H00269	EDU	22.0	65.0	66.8	78.0	231.8	60.75

Data Projection

```
Assign grades for each student based on final marks:
 75 & above - A
60 to 74 - B
35 to 59 - C
Below 35 - F
'''
# Defining all the conditions inside a function
def condition(x):
    if x>=75:
        return "A"
    elif x>=60 and x<=74:
        return "B"
    elif x>=35 and x<=59:
        return "C"
    else:
        return 'F'

# Applying the conditions
grades['Grade'] = grades['Final Marks'].apply(condition)
print(grades)
```

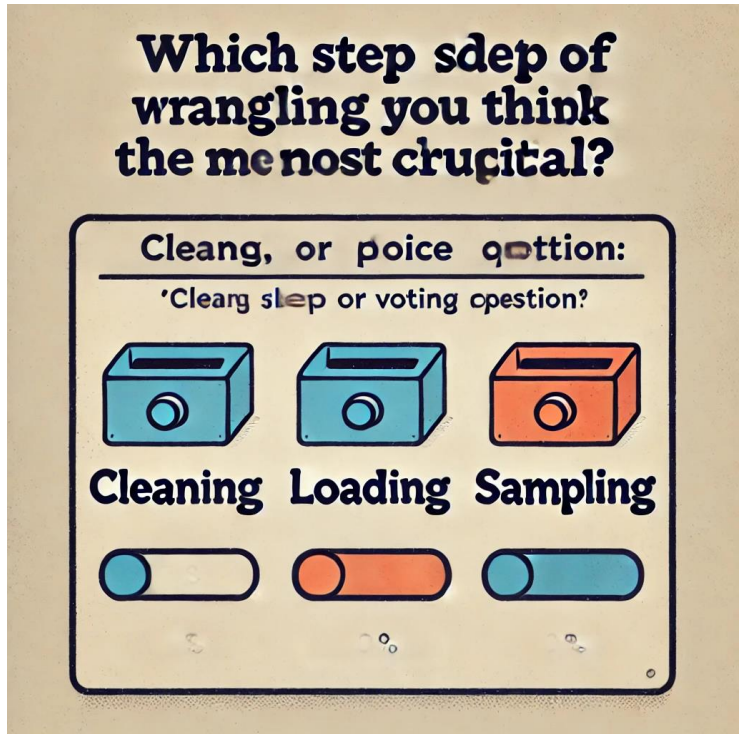
	ID	Division	Quiz	Practical	Project	FWA	Total	Final Marks	Grade
0	H00247	EDU	84.0	67.0	69.6	67.0	287.6	71.05	B
1	H00258	HLT	74.0	68.0	67.0	73.0	282.0	70.45	B
2	H00269	ENG	88.0	86.0	75.6	74.0	323.6	80.20	A
3	H00224	EDU	65.4	71.0	65.8	71.0	273.2	68.58	B
4	H00269	EDU	22.0	65.0	66.8	78.0	231.8	60.75	B

Data Projection

```
'''
Write the following command to write data into a excel file
• Create a excel file grades.xlsx
• Write the data from data frame grades to worksheet Grades_Sheet
• Save the file
'''
writer = pd.ExcelWriter('grades.xlsx')
grades.to_excel(writer,'Grades_Sheet')
writer.save()
```

Thinks to Ponder

Which step of data wrangling do you think is the most crucial: Cleaning, Loading, or Sampling?



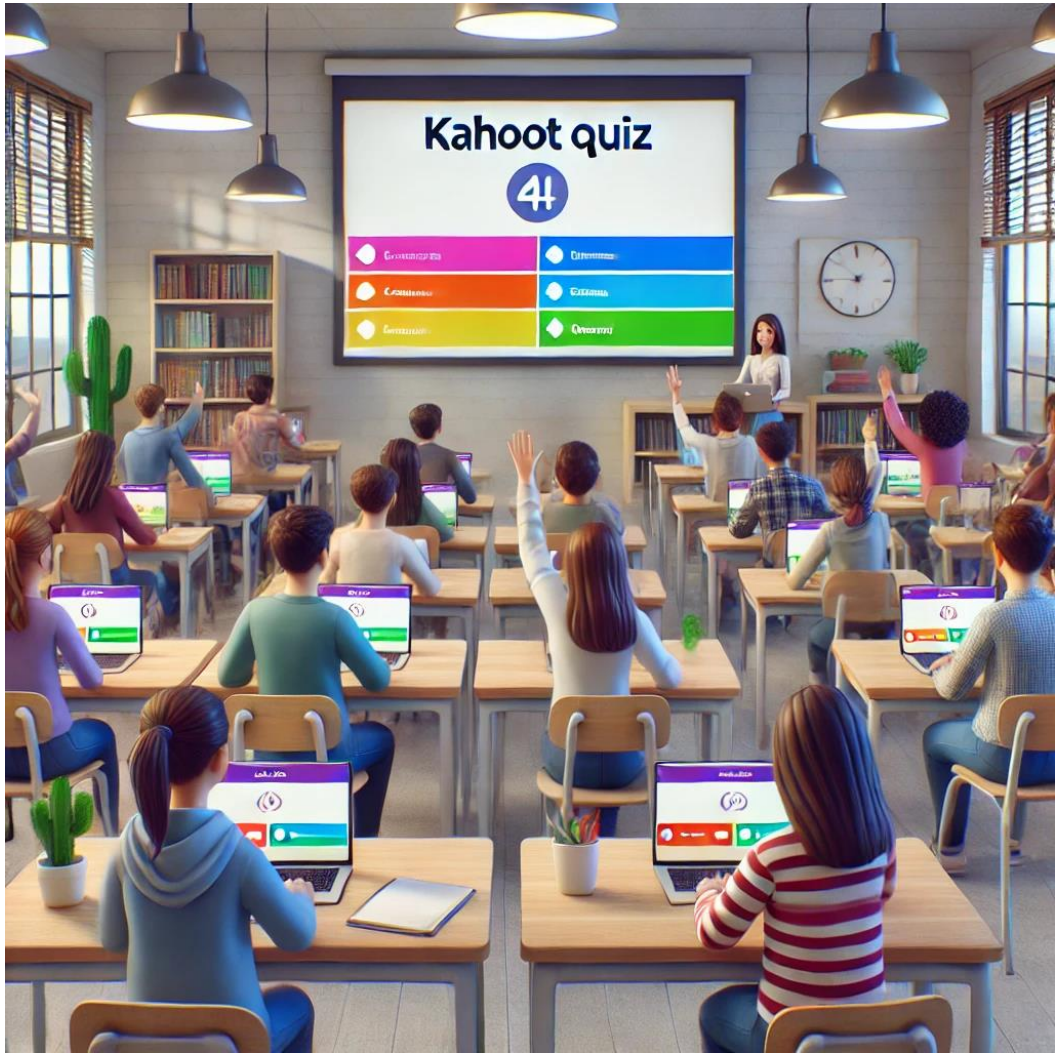
Summary

- Importance of Data Wrangling:
  - Improves data quality by fixing errors and handling missing values.
  - Increases usability for analysis, supporting better decision-making.
- Steps in Data Wrangling:
  - Data Loading: Importing data using Python libraries like Pandas.
  - Data Cleaning: Techniques to address missing or inconsistent data.
  - Data Sampling: Using random and systematic methods to reduce data size.
  - Data Projection: Focusing on relevant data columns for efficient analysis.

Summary

- Practical Examples:
  - Real-world datasets were used to demonstrate Python code for cleaning and sampling data.

Kahoot Time!



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Thank You